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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603 203

SET – 1

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16-02-23
Duration: 08:00 – 08:50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. The crystal field splitting energy for octahedral and tetrahedral complexes is related as
a) $\Delta_t = 4/9\ \Delta_o$

2. Which of the following octahedral complexes of Co (atomic no.27) will have the highest Δ_o value?
a) $[Co(CN)_6]^{3-}$

3. Which of the following species will be diamagnetic?
a) $[Fe(CN)_6]^{4-}$
[Note: Even though $C_2O_4^{2-}$ is a weak field ligand, it acts as a strong field ligand with Co^{3+}. Marks can be given for option c) $[Co(C_2O_4)_3]^{3-}$ answer also.]

4. The screening constant of 2s electron in lithium atom is_____
d) 1.70

5. According to HSAB concept, which of the following will prefer to occur as oxides
d) Al^{3+}

Part – B (2 x 10 = 20 Marks)

6. a. i Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why? (4 Marks)

(A). $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$

Ans: $[Co(H_2O)_6]^{2+}$ - 1 Mark

Reason: Co^{2+} - $3d^7$ and Ni^{2+} - $3d^8$ and No. of d-electrons α $1/\Delta_o$ - 1 Mark

(B) $[V(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{3+}$

Ans: $[Cr(H_2O)_6]^{3+}$ – 1 Mark

Reason: Higher the oxidation state of the cation, larger the Δ_o value – 1 Mark

ii. Calculate the CFSE and magnetic moment values for the high and low spin octahedral complexes of Co(II) [Z = 27] in terms of Δ_o and P. Predict whether the

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complexes are paramagnetic and diamagnetic. (6 Marks)

L.S complex of Co(II) – CFSE + Magnetic moment calculations – 2 + 1 Marks

H.S complex of Co(II) – CFSE + Magnetic moment calculations – 2 + 1 Marks

(OR)

b. i. Discuss any two factors that influence the magnitude of crystal field splitting, Δ_o. (6 Marks)

Any two factors with example: 2 x 3 Marks (each) = 6 Marks

ii. $[Ti(H_2O)_6]^{3+}$ is coloured, while $[Sc(H_2O)_6]^{3+}$ is colourless – Explain. [Note: Atomic number of Sc and Ti is 21 and 22 respectively]. (4 Marks)

Ti^{3+} - $3d^1$ configuration; d – d transition is possible. So, coloured complex. – 2 Marks

Sc^{3+} - $3d^0$ configuration; d – d transition is not possible. So, colourless complex. –2 Marks

7 a. i Elaborate the ionization and coordination isomerism shown by the octahedral complexes with an example. (6 Marks)

Ionization isomerism: definition with an example – 3 Marks

Coordination isomerism: definition with an example – 3 Marks

ii. Calculate the effective nuclear charge of a 3p electron in the chlorine atom. (Number of Protons = 17). (4 Marks)

Ans: Z_{eff} of Cl = 17-10.9 = 6.1

(OR)

b. i. Define ionization potential? Give its variation along the period and down the group with examples. (6 Marks)

Definition – 1 Mark

Explanation with an exemption – 5 Marks

ii. Write a note on the application of HSAB principle in explaining the stability of complexes with the examples. (4 Marks)

Explanation by taking any two examples – 4 Marks

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Set-2

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech	Date: 16.02.23
Course Code & Title: 21CYB101J & Chemistry	Duration: 8.00-8.50 AM
Year & Sem: I Year & II Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 Crystal field stabilization energy for high spin d^4 octahedral complex is:

d) $-0.6\ \Delta_0$

2 The largest paramagnetism is shown by _____

(b) $[Fe(H_2O)_6]^{2+}$

3 What kind of isomerism(s) is(are) possible for a complex $[Co(NH_3)_4\ Br_2]Cl$?

a) geometrical and ionization

4 Choose the correct order of ionization energy:

b) N > O < F

5 Which of the following will prefer to exist as sulphide?

a) Hg^{2+}

Part – B (2 x 10 = 20 Marks)

6. a. Write a short note on spectrochemical series? Explain with a suitable example, the influence of ligands in the formation of high and low spin complexes. (10 Marks)

Definition and series of ligands – 4 Marks

Splitting diagram of HS and LS complex with explanation of an example–6Marks

(OR)

b i. Arrange the following complexes in their increasing order of wavelength of light absorbed and explain. (6 Marks)
$[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$ and $[Co(H_2O)_6]^{3+}$

Δ_0 α Splitting energy α 1/wavelength, λ
Stronger the ligand more the splitting; So, the increasing order of λ absorbed is $[Co(CN)_6]^{3-} < [Co(NH_3)_6]^{3+} < [Co(H_2O)_6]^{3+}$

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ii. Calculate the crystal field splitting energy, Δ_o for $[Cu(H_2O)_6]^{2+}$ complex at the maximum absorbance (λ_{max}) of 600nm and predict its colour. (4 Marks)

$\Delta_0 = hc/\lambda = (6.626 \times 10^{-34}\ J.s)(3 \times 10^8 m/s)/(600 \times 10^{-9} m) = 3.313 \times 10^{-19} J$ – 3 Marks

Color of the complex observed with the absorption of λ_{max}=600nm is blue–1Mark

7. a. i. Discuss the hydration and linkage isomerism shown by the octahedral complexes with an example. (6 Marks)

Hydration isomerism: definition with an example – 3 Marks

Linkage isomerism: definition with an example – 3 Marks

ii. Determine the screening constant and effective nuclear charge of 2p electron in Mg^{2+} ion. [Number of protons = 12]. (4 Marks)

Ans: Z_{eff} of Mg^{2+} ($1s^2\ 2s^2\ 2p^6$) = 12-4.15 = 7.85

(OR)

b. i. Define atomic and ionic radii. Arrange the following O, O^- and O^{2-} in their increasing order of radii. (6 Marks)

Definition – 2 Marks

Order of arrangement: $O < O^- < O^{2-}$ – 1 Mark

Explanation – 3 Marks

ii. According to HSAB principle, predict whether the following reaction is feasible or not.

$$BeI_2 + HgF_2 \rightarrow BeF_2 + HgI_2$$ (4 Marks)

Ans: Reaction is feasible – 1 Mark

Explanation – 3 Marks

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Set-1

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16.02.23
Duration: 12.30 – 1.20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is
   a) $[Fe(H_2O)_6]^{3+}$

2. How many unpaired electrons are there in a strong field iron(II) octahedral complex?
   a) 0

3. Which one of the following complexes is NOT expected to exhibit isomerism?

   b) $[Ni(NH_3)_2Cl_2]$

4. Choose the correct statement
   a) As shielding effect increases electronegativity decreases

5. Which of the following transition metal ions will prefer to form thiocyanato (SCN^-) complex on the basis of HSAB principle?
   d) Pt^{4+}

Part – B (2 x 10 = 20 Marks)

6. a. i. Compare the CFSE and magnetic moment values of $[Ni(CN)_6]^{4-}$ and $[Ni(H_2O)_6]^{2+}$ complexes. [Atomic number of Ni = 28] (6 Marks)

$[Ni(CN)_6]^{4-}$ - L.S complex–CFSE + Magnetic moment calculations–2 Marks
$[Ni(H_2O)_6]^{2+}$ - H.S complex–CFSE + Magnetic moment calculations–2 Marks
Comparison and discussion – 2 Marks

ii. Explain why the tetrahedral complexes are always high spin? (4 Marks)

1) d-orbitals are not pointing directly towards the ligands to have high splitting energy
2) Less number of ligands (CN is 4) leading to low splitting energy
3) Thus, $\Delta_t < P$ (pairing energy)
4) So, pairing of electrons is not energetically favourable

(OR)

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b. Explain in detail the d-orbital splitting in octahedral complexes and d-d transition with a suitable example. (10 Marks)

Crystal Field Splitting Diagram- 2 Marks,

Explanation on d – orbitals splitting – 4 Marks,

Describing d-d transition with example – 4 Marks

6. a. i. Brief about the geometrical isomerism exhibited by octahedral complexes. (6 Marks)

Definition – 2 Marks

Discussion with any two examples – 4 Marks

ii. Calculate the effective nuclear charge experienced by the valence electron of Zn [Z=30]. (4 Marks)

Ans: Zeff = 30 – [(18x1) + (9x0.35)] = 30 – 21.15 = 8.85

(OR)

b. i. Define the term Electron affinity. Discuss its variation across the period and down the group with examples. (6 Marks)

Definition – 1 Mark

Explanation with an exemption – 5 Marks

ii. Aluminium occurs in nature as oxide whereas Platinum occurs as sulphide. Explain based on HSAB principle. (4 Marks)

Aluminium – oxide (Hard acid – Hard base) – Explanation - 2 Marks

Platinum – sulphide (Soft acid – Soft base) – Explanation - 2 Marks

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SET – 2

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16-02-23
Duration: 12.30-1.20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. A magnetic moment of 1.73BM will be shown by one among the following
   c) $[Cu(NH_3)_4]^{2+}$

2. The ligand that produces the strongest field and causes maximum splitting is _____
   d) CO

3. Which type of isomerism is exhibited by $[Pt(NH_3)_2Cl_2]$?
   d) Geometrical isomerism

4. Which of the following species has the highest ionization potential?
   a) Li^+

5. According to HSAB concept, which of the following will prefer to exist as carbonates
   b) Mg^{2+}

Part – B (2 x 10 = 20 Marks)

6. a. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[Mn(Cl_4)]^{2-}$ (Atomic number of Mn is 25). (10 Marks)

Crystal Field Splitting Diagram- 2 Marks,
Explanation on d – orbitals splitting – 4 Marks,
CFSE calculation – 2 Marks
Magnetic moment calculation – 2 Marks

(OR)

b. i. Describe the salient features of Crystal Field Theory. (4 Marks)

Features of CFT - 4 points – 4 Marks

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SET – 2

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 16-02-23
Duration: 12.30-1.20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. A magnetic moment of 1.73BM will be shown by one among the following
   c) $[Cu(NH_3)_4]^{2+}$

2. The ligand that produces the strongest field and causes maximum splitting is _____
   d) CO

3. Which type of isomerism is exhibited by $[Pt(NH_3)_2Cl_2]$?
   d) Geometrical isomerism

4. Which of the following species has the highest ionization potential?
   a) Li^+

5. According to HSAB concept, which of the following will prefer to exist as carbonates
   b) Mg^{2+}

Part – B (2 x 10 = 20 Marks)

6. a. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[Mn(Cl_4)]^{2-}$ (Atomic number of Mn is 25). (10 Marks)

Crystal Field Splitting Diagram- 2 Marks,
Explanation on d – orbitals splitting – 4 Marks,
CFSE calculation – 2 Marks
Magnetic moment calculation – 2 Marks

(OR)

b. i. Describe the salient features of Crystal Field Theory. (4 Marks)

Features of CFT - 4 points – 4 Marks

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ii An octahedral complex absorbs light with a wavenumber of 18,700 cm^{-1}. Determine the crystal field splitting energy, Δ_o and complementary colour of the complex? (6 Marks)

$\Delta_o = hc\bar{\nu} = (6.626 \times 10^{-34}\ J.s)(3 \times 10^{10}\ cm/s)(18700\ cm^{-1}) = 3.717 \times 10^{-19}\ J$ - 4 Marks

λ= 1/ Wavenumber = 5.348×10^{-5} cm = 534.8 nm - 1 Mark

Colour of the complex observed is Red (or violet) – 1 Mark

7 a. i Discuss the optical isomerism exhibited by the transition metal compounds with any two examples. (6 Marks)

Definition – 2 Marks
Discussion with any two examples – 4 Marks

ii. What is the effective nuclear charge experienced by a 3d electron in Cu [Z=29]? (4 Marks)

Slater configuration of Cu: $(1s^2)\ (2s^2)\ (2p^6)\ (3s^2)\ (3p^6)\ (3d^{10})\ (4s^1)$ - 1 Mark

Effective nuclear charge = 29 – [(18x1) + (9x0.35)] = 7.85 - 3 Marks

(OR)

b. i. Define Electronegativity? Give its variation along the period and down the group with examples. (6 Marks)

Definition – 1 Mark
Explanation with an exemption – 5 Marks

ii. Give a brief account on hard and soft acids. (4 Marks)

Features of
HA with examples – 2 Marks
SA with examples – 2 Marks

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SET – I

ANSWER-KEY

INTERNAL ASSESSMENT – I

Program: B.Tech	**Date: 13/02/2023**
Course Code & Title:21CYB101J & Chemistry	**Duration: 8.00 – 8.50 AM**
Year & Sem: I Year & II Sem	**Max. Marks: 25 marks**

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. Which of the following complex is paramagnetic? (Z of Cu = 29, Fe = 26, Co = 27)
   (c) $[CoBr]_4^{2}$
2. The correct increasing order of splitting power of ligands according to spectrochemical series is
   (a) $Cl^- < OH^- < CN^-$
3. Which one of the following ions exhibits colour in aqueous solution?
   (b) Ni^{2+}
4. Which of the following order is correct for the first ionization energies of following elements?
   (c) B < Be < O < N
5. According to HSAB theory, which option shows negative correlation with hardness?
   (b) Polarizability

Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals, calculation of CFSE and magnetic moment in case of tetrahedral complexes? (10 Marks)

Crystal Field Splitting Diagram- 3 Marks,
Explanation – 3 Marks,
CFSE formula – 2 marks
Magnetic moment – 2 marks

(OR)

b. i. For Mn^{2+} ion, P = 28,000 cm^{-1} and Δ_o value for $[Mn(CN)_6]^{3-}$ ion is 38,500 cm^{-1}. Predict whether the complex is high spin or low spin. Write the configuration and calculate CFSE in cm^{-1}.

Ans: Low spin , CFSE = -33600 cm^{-1}

$[Mn(CN)_6]^{3-}$ has $t_{2g}^4 e_g^0$ (6 Marks)

$CFSE = -\frac{2}{5} \times 4 \times 38500 + 28000 \times 1 = -33,600\ cm^{-1}$

(P)

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ii. Write the formula for ionization isomer of $[Co(en)_2Cl.SCN]NO_2$ and hydration isomer of $[Cr(H_2O)_6]Cl_3$ (4 Marks)

Ans: Ionization isomers – 2 marks

Hydration isomers – 2 marks

Formula explanation:

$[Co(en)_2Cl(NO_2)]SCN$

(i) $[Cr(H_2O)_6]Cl_3$ - violet

(ii) $[Cr(H_2O)_5]Cl_2 \cdot H_2O$ - A pale green

(iii) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ dark green color

7 a. Explain HSAB principle. Give its applications and limitations. (10 Marks)

Ans: Features of HA, SA, HB, SB, HSAB principle with explanation – 6 marks

Two applications – 2 marks

Two limitations – 2 marks

(OR)

b. i. Calculate the octahedral crystal field splitting energy in kJ for $[Fe(CN)_6]^{4-}$, if the wavelength of the most intensely absorbed light is 305 nm.
($h = 6.626 \times 10^{-34}$ Js, $c = 2.99 \times 10^8$ m/s) (6 Marks)

Ans: 6.513×10^{-22} kJ or 392 kJ/mol

ii. Atomic size decreases from Li to F in a second period, however from F to Ne it increases, explain. (4 Marks)

Ans: Explanation for decrease in size from Li to F – 2 marks

Explanation for increase from F to Ne (mention of atomic and Van der Waals radius) – 2 marks

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SET – II

ANSWER-KEY

INTERNAL ASSESSMENT – I

Program: B.Tech	Date: 13/02/2023
Course Code & Title:21CYB101J & Chemistry	Duration: 8.00 – 8.50 AM
Year & Sem: I Year & II Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1. Which of the following complexes has a magnetic moment of 1.73 BM?
(Z of Ni = 28, Ti = 22, Cu = 29, Co = 27)
(c) $[Cu(NH_3)_4]^{2+}$

2. The correct electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory is
(c) $t_{2g}^6\ e_g^0$

3. Out of TiF_6^{2-}, CoF_6^{3-}, $Fe(CN)_6^{4-}$ and $NiCl_4^{2-}$ (Z of Ti = 22, Co = 27, Fe = 26, Ni = 28), the colourless species is :
(a) TiF_6^{2-}

4. The electron affinity for the inert gases is
(a) 0

5. According to hard and soft acid base principle, hard acid
(a) is not polarizable

Part – B (2 x 10 = 20 Marks)

6. a. Describe structural isomerism in co-ordination compounds with suitable examples. (10 Marks)

Ans: Ionization isomerism, Hydrate isomerism, Linkage isomerism, Coordination isomerism –explanation with example – each 2.5 marks

(OR)

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b. i. Calculate CFSE values in terms of Δ_o and P for high and low spin octahedral complexes of Fe (II). Predict whether the complexes are paramagnetic or diamagnetic? (4 Marks)

Ans: High spin: -0.4 Δ_o + P, paramagnetic – 2 marks
Low spin: -2.4 Δ_o + 3P, diamagnetic – 2 marks

ii. Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why?

(a) $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b) $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$

(c) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$ (6 Marks)

Ans: (a) $[Fe(H_2O)_6]^{3+}$- Higher oxidation state – 2 marks

(b) $[Co(H_2O)_6]^{2+}$ - lower number of d electrons – 2 marks

(c) $[Rh(NH_3)_6]^{3+}$ - higher quantum number of d orbital of metal centre – 2 marks

7 a. i Define 'Ionization Energy', discuss factors affecting it and its variation across the period & down the group in a periodic table. (10 Marks)

Ans: Definition – 2 marks, any four factors – 4 marks, variation across period – 2 marks, variation down the group – 2 marks

(OR)

b. i. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of V (Z= 23) (6 Marks)

Ans: 4.30

Electronic configuration – 1 marks, Formula for Z_{eff} – 1 mark, calculation of screening constant – 3 marks, calculation of Z_{eff} – 2 marks.

ii. Explain complex formation of SCN^- with Pt^{2+} and Cr^{3+}. (4 Marks)

Ans: Explanation based on HSAB principle – 4 marks

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SET – 1

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13-02-23
Duration: 8.00 – 8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Which of the following complex is paramagnetic? (Z of Cu = 29, Fe = 26, Co = 27)

   (a) $[Cu(CN)_4]^{3-}$ (b) $[Fe(CN)_6]^{4-}$ (c) $[CoBr]_4^2$ (d) $[Co(NH_3)_6]^{3+}$

2. The correct increasing order of splitting power of ligands according to spectrochemical series is

   (a) $Cl^- < OH^- < CN^-$ (b) $Cl^- < CN^- < OH^-$
   (c) $OH^- < Cl^- < CN^-$ (d) $OH^- < CN^- < Cl^-$

3. Which one of the following ions exhibits colour in aqueous solution?

   (a) Sc^{3+} (b) Ni^{2+} (c) Ti^{4+} (d) Zn^{2+}

4. Which of the following order is correct for the first ionization energies of following elements?

   (a) B < Be < N < O (b) Be < B < N < O
   (c) B < Be < O < N (d) B < O < Be < N

5. According to HSAB theory, which option shows negative correlation with hardness?

   (a) Oxidation State (b) Polarizability
   (c) Electronegativity (d) Charge density

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Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals, the calculation of CFSE and magnetic moment in the case of tetrahedral complexes? (10 Marks)

(OR)

b. i. For Mn^{2+} ion, P = 28,000 cm^{-1} and Δ_o value for $[Mn(CN)_6]^{3-}$ ion is 38,500 cm^{-1}. Predict whether the complex is high spin or low spin. Write the configuration and calculate CFSE in cm^{-1}. (6 Marks)

ii. Write the formula for ionization isomer of $[Co(en)_2Cl.SCN]NO_2$ and hydration isomer of $[Cr(H_2O)_6]Cl_3$ (4 Marks)

7 a. Explain HSAB principle. Give its applications and limitations. (10 Marks)

(OR)

b. i. Calculate the octahedral crystal field splitting energy in kJ for $[Fe(CN)_6]^{4-}$, if the wavelength of the most intensely absorbed light is 305 nm.
($h = 6.626 \times 10^{-34}$ Js, $c = 2.99 \times 10^8$ m/s) (6 Marks)

ii. Atomic size decreases from Li to F in a second period; however from F to Ne it increases. Give reasons. (4 Marks)

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SET – 2

INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13-02-23
Duration: 8.00 – 8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Which of the following complexes has a magnetic moment of 1.73 BM? (Z of Ni = 28, Ti = 22, Cu = 29, Co = 27)

(a) $[Ni(CN)_4]^{2-}$ (b) $TiCl_4$ (c) $[Cu(NH_3)_4]^{2+}$ (d) $[CoCl_6]^{4-}$

2. The correct electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory is

(a) $e^4 t_2^2$ (b) $t_{2g}^4 e_g^2$ (c) $t_{2g}^6 e_g^0$ (d) $e_1 t_2^3$

3. Out of TiF_6^{2-}, CoF_6^{3-}, $Fe(CN)_6^{4-}$ and $NiCl_4^{2-}$ (Z of Ti = 22, Co = 27, Fe = 26, Ni = 28), the colourless species is :

(a) TiF_6^{2-} (b) CoF_6^{3-} (c) $Fe(CN)_6^{4-}$ (d) $NiCl_4^{2-}$

4. The electron affinity for the inert gases is

(a) 0 (b) High (c) Negative (d) Positive

5. According to hard and soft acid base principle, hard acid

(a) is not polarizable
(b) has low charge density
(c) shows preference for donor atoms of low electronegativity
(d) shows preference for soft bases

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Part – B (2 x 10 = 20 Marks)

a. Describe structural isomerism in co-ordination compounds with suitable examples. (10 Marks)

(OR)

b. i. Calculate CFSE values in terms of Δ_o and P for high and low spin octahed complexes of Fe (II). Predict whether the complexes are paramagnetic diamagnetic? (4 Mark

ii. Which of the following complexes have large crystal field splitting of d-orbitals each pair and why?

(a) $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b) $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$

(c) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$ (6 Mar

Ionization Energy: Discuss the factors affecting it and its variation ac (10 Mar

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Part – B (2 x 10 = 20 Marks)

6. a. Describe structural isomerism in co-ordination compounds with suitable examples. (10 Marks)

(OR)

b. i. Calculate CFSE values in terms of Δ_o and P for high and low spin octahedral complexes of Fe (II). Predict whether the complexes are paramagnetic or diamagnetic? (4 Marks)

ii. Which of the following complexes have large crystal field splitting of d-orbitals in each pair and why?

(a) $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$ (b) $[Co(H_2O)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$

(c) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$ (6 Marks)

7 a. i Define 'Ionization Energy'. Discuss the factors affecting it and its variation across the period and down the group in a periodic table. (10 Marks)

(OR)

b. i. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of V (Z= 23) (6 Marks)

ii. Explain complex formation of SCN^- with Pt^{2+} and Cr^{3+}. (4 Marks)

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DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set-1

Answer -Key
INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13/02/2023
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 $[Cr(CN)_6]^{3-}$ is expected to be: (Z of Cr = 24)

(a) paramagnetic with $\mu_{eff} \approx 3.87$ BM

2 For which pair of complexes is the order of values of Δ_{oct} correct?

(a) $[Rh(NH_3)_6]^{3+} > [Co(NH_3)_6]^{3+}$

3 What is the secondary valency of metal ion in $[Cr(NH_3)_4(en)]NO_3$?

(c) 6

4 Which of the following is the correct order of elements with increasing electronegativity?

(d) P < S < N < O

5 Which of the following metals are usually found in the form of oxide ores?

(b) Fe^{3+} and Al^{3+}

Part – B (2 x 10 = 20 Marks)

6. a. Explain crystal field splitting of d orbitals, calculation of CFSE and magnetic moment in case of octahedral complexes? (10 Marks)

Ans: Crystal Field Splitting Diagram- 3 Marks,
Explanation – 3 Marks,
CFSE formula – 2 marks
Magnetic moment – 2 marks

(OR)

b. i. For $[Cr(H_20)_6]^{2+}$ ion the mean pairing energy (P) is found to be equal to 23500 cm^{-1}. The magnitude of Δ_o is 13900 cm^{-1}. Calculate the CFSE for this complex ion corresponding to high spin and low spin state. (6 Marks)

Ans: CFSE for HS state = -8340 cm^{-1}
CFSE for LS state = + 1260 cm^{-1}

[N.B: steps must be given]

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ii. Draw structures for linkage isomers of $[FeCl_5SCN]^{3-}$ and optical isomers of $[PtCl_2(en)_2]^{2+}$

Ans: Linkage isomers – 2 marks
Optical isomers – 2 marks
(Only the structures) (4 Marks)

7 a. i Explain HSAB principle. Give its applications and limitations. (10 Marks)

Ans: Features of HA, SA, HB, SB, HSAB principle with explanation – 6 marks
Two applications – 2 marks
Two limitations – 2 marks

(OR)

b. i. The octahedral complex $[Ti(H_2O)_6]^{3+}$ has a single d electron. To excite this electron from the ground state t_{2g} orbital to the eg orbital, this complex absorbs light from 450 to 600 nm. The maximum absorbance corresponds to Δ_o and occurs at 499 nm. Calculate the value of Δ_o in Joules and predict what color the solution will appear. ($h = 6.626 \times 10^{-34}$ Js, $c = 2.99 \times 10^8$ m/s)

Ans: E = 3.99 x 10-19 J – 4 marks (6 Marks)

Because the complex absorbs 600 nm (orange) through (blue), the indigo, violet and red wavelengths will be transmitted and the complex will appear purple. – 2 marks

ii. In a second period the ionization energy is expected to vary in following order Li < Be < B < C < N < O < F < Ne, however the observed trend is Li < B < Be < C < O < N < F < Ne, explain. (4 Marks)

Ans: Easier to remove electron from B ($2s^2$, $2p^1$) compared to Be ($2s^2$) thus IE of B is less than Be – 2 marks
Easier to remove electron from O ($2s^2$, $2p^4$) compared to N ($2s^2$, $2p^3$) thus IE of O is less than N – 2 marks

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Set-2

Answer -Key
INTERNAL ASSESSMENT – I

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 13/02/2023
Duration: 12:30 – 1:20 PM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)

Answer ALL The Questions

1 The co-ordination number, oxidation number, number of electron in d-orbital and number of unpaired electrons in complex $[Co(en)_2(NH_3)_2]^{2+}$ is respectively
(c) 6,2,7,1

2 What is theoretical magnetic moment of complex compound $K_2[NiF_4]$?
(b) 2.83 BM

3 Among SCN^-, NO_3^-, NO_2^- and $C_2O_4^{2-}$ which are ambidentate ligands?
(a) SCN^- and NO_2^-

4 Which is smallest in size
(c) F^-

5 According to hard and soft acid base principle, a hard acid:
(d) is not polarizable

Part – B (2 x 10 = 20 Marks)

6. a. i. Describe stereo-isomerism in co-ordination compounds with suitable examples. (10 Marks)

Ans: Explanation of geometric isomerism in square planar and octahedral geometry with examples – 2 marks
Explanation of optical isomerism with examples – 2 marks

(OR)

b. i. Ni^{2+} in a coordination complex is paramagnetic. Draw the crystal field splitting diagrams for an octahedral crystal field and a tetrahedral crystal field, labeling the d-orbitals. State below each diagram whether the geometry is consistent with a paramagnetic species. (6 Marks)

Ni^{2+} - (d^8,Oh) – Diagram- paramagnetic (2 unpaired electrons) – 3 marks
(d^8, Td) - Diagram- paramagnetic (2 unpaired electrons) – 3 marks

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ii. The strong ligand complexes are usually yellow, orange or red in color while weak ligand field complexes are blue, green or indigo in color, explain. (4 Marks)

Ans: Strong ligand complexes absorb high energy violet or blue light therefore they are usually yellow, orange or red in color, on the other hand, weak ligand complexes absorb lower energy yellow orange or red light and hence are blue, green or indigo in color. – explanation based on extent of crystal field splitting – 4 marks

7 a. Define and explain the following periodic properties :

(A) Van der Waals radius (B) Electron affinity

(C) Ionization energy (D) Electronegativity

Ans: Definition and explanation – each 2.5 marks (10 Marks)

(OR)

b. i. Calculate Effective Nuclear Charge (Z_{eff}) for 3d electron of Cr (Z= 24)

Ans: 4.60

Electronic configuration – 1 marks, Formula for Z_{eff} – 1 mark, calculation of screening constant – 3 marks, calculation of Z_{eff} – 2 marks. (6 Marks)

ii. Predict and explain the products of the following reaction:

LiI + CsF ---------- ? LiF + CsI

Ans: Explanation based on HSAB principle – 4 marks (4 Marks)

Ans : LiF + CsI

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